

SQL BUSINESS ANALYSIS

Save for Later Feature — SQL Business Analysis

Business Questions, Query Design & Data-Driven Recommendations

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1. Document Control

1.1 Revision History

Version	Date	Author	Changes
1.0	21st July 2026	Aayush Saini	Initial set of 14 business questions covering customer, product, revenue and conversion analysis.
2.0	6th August 2026	Aayush Saini	Corrected Revenue Per Customer to align with Power BI dashboard definition; added 5 new questions covering monthly trend, day-of-week pattern, month-over-month growth, running totals, and a correlated subquery; converted all results to formatted tables; completed all previously truncated result sets.

2. Purpose of This Analysis

This document translates the business and functional requirements defined in the Save for Later BRD and FRD into a structured set of SQL business questions. Each question follows a consistent format — objective, query, result, key KPIs, business insight, recommendation, and business value — so the analysis reads as a decision-support document, not just a collection of queries.

Version 2.0 corrects a metric-definition inconsistency identified during QA (Revenue Per Customer), completes several result sets that were previously truncated, and adds five new questions covering time-based trend analysis, window functions, and a correlated subquery — extending the technical range demonstrated beyond the original v1.0 scope.

2.2 Data Sources

Table	Rows	Key Fields
Customers	500 rows	Customer_ID, Customer_Name, City, Gender, Age_Group, Customer_Type, Registration_Date
Orders	1,289 rows	Order_ID, Customer_ID, Product_ID, Order_Date, Order_Value, Payment_Method, Order_Status, Discount
Products	96 rows	Product_ID, Product_Name, Category, Brand, Price, Stock_Status
Save_For_Later	3,000 rows	Save_ID, Customer_ID, Product_ID, List_Name, Save_Date, Purchase_Status, Purchase_Date, Days_To_Purchase

2.3 Assumptions & Data Notes

- All datasets are synthetically generated for portfolio purposes and are not sourced from a live production system.
- Order_Date values fall between June and mid-December 2026; December is a partial month and is called out explicitly wherever it would otherwise distort a trend.
- Of the 500 registered customers, 456 (91.2%) have placed at least one order; 44 customers have never ordered and are excluded from any query that joins through Orders.
- “Revenue Per Customer” is standardized throughout this document to Total Revenue ÷ all 500 registered customers, to match the definition used in the Power BI dashboard. An earlier version of this document (v1.0) divided by ordering customers only (456), producing ₹1,180.71 instead of ₹1,076.81 — that inconsistency is corrected here.
- Order_Value is treated as gross revenue; the Discount field is not netted out, consistent with how Total Revenue is reported on the Power BI Executive Dashboard.

3. Customer & Geographic Analysis

3.1 — Which cities have the highest number of registered customers?

OBJECTIVE

Identify geographical distribution of customers to support marketing and acquisition strategy.

SQL QUERY

```
SELECT City,  
       COUNT(*) AS Total_Customers  
FROM Customers  
GROUP BY City  
ORDER BY Total_Customers DESC;
```

RESULT

City	Total Customers
Ghaziabad	58
Faridabad	57
Pune	55
Chandigarh	53
Bengaluru	50
Jaipur	49
Delhi	49
Lucknow	47
Gurgaon	42
Noida	40

KEY KPIS

- Largest customer base: Ghaziabad (58)
- Smallest customer base: Noida (40)
- Top 3 cities hold 34% of all registered customers

BUSINESS INSIGHT

Customer registration is concentrated in Ghaziabad, Faridabad and Pune, which together account for over a third of the base. Noida and Gurgaon have the smallest registered populations, despite Gurgaon posting a strong average order value elsewhere in this analysis.

RECOMMENDATION

- Increase acquisition spend in underperforming cities such as Noida and Gurgaon.
- Maintain investment in high-registration cities to protect market share.
- Cross-check registration volume against revenue per city (Q6) before reallocating budget — a large customer base does not guarantee proportional revenue.

BUSINESS VALUE

Geographic distribution analysis helps marketing allocate acquisition budget toward the regions most likely to grow the customer base efficiently.

3.2 — Which cities generate the highest revenue?

OBJECTIVE

Identify the highest revenue-generating cities to support regional sales and marketing decisions.

SQL QUERY

```
SELECT c.City,
       COUNT(*) AS Total_Orders,
       ROUND(SUM(o.Order_Value), 2) AS Total_Revenue,
       ROUND(AVG(o.Order_Value), 2) AS Avg_Order_Value
FROM Customers AS c
INNER JOIN Orders AS o
      ON c.Customer_ID = o.Customer_ID
GROUP BY c.City
ORDER BY Total_Revenue DESC;
```

RESULT

City	Total Orders	Total Revenue	Avg Order Value
Ghaziabad	160	₹65,961.10	₹412.26
Faridabad	140	₹59,460.05	₹424.71
Jaipur	137	₹56,969.00	₹415.83
Chandigarh	136	₹56,730.70	₹417.14
Pune	132	₹55,319.80	₹419.09
Bengaluru	123	₹54,503.25	₹443.12
Delhi	124	₹51,930.80	₹418.80
Gurgaon	114	₹51,116.45	₹448.39
Lucknow	114	₹46,364.95	₹406.71
Noida	109	₹40,048.25	₹367.42

KEY KPIS

- **Highest revenue: Ghaziabad (₹65,961.10)**
- **Highest average order value: Gurgaon (₹448.39)**
- **Lowest average order value: Noida (₹367.42)**

BUSINESS INSIGHT

Ghaziabad leads on total revenue mainly through order volume, while Gurgaon customers spend more per order despite a smaller base. Noida is weakest on both order volume and average order value, making it the clearest underperformer of the ten cities.

RECOMMENDATION

- Continue investing in Ghaziabad to protect the largest revenue base.
- Study Gurgaon's higher average order value to identify transferable pricing or merchandising patterns.
- Prioritize Noida for a targeted spend-recovery campaign rather than acquisition alone, since its issue is order value, not customer count.

BUSINESS VALUE

Revenue-by-city analysis lets management identify the platform's most profitable regions and build region-specific growth plans instead of a one-size-fits-all rollout.

3.3 — Which customer segment (New vs Returning) generates more revenue?

OBJECTIVE

Compare revenue contribution between new and returning customers to evaluate retention performance.

SQL QUERY

```
SELECT c.Customer_Type,
       COUNT(DISTINCT c.Customer_ID) AS Total_Customers,
       COUNT(*) AS Total_Orders,
       ROUND(SUM(o.Order_Value), 2) AS Total_Revenue,
       ROUND(AVG(o.Order_Value), 2) AS Avg_Order_Value
FROM Customers AS c
INNER JOIN Orders AS o
      ON c.Customer_ID = o.Customer_ID
GROUP BY c.Customer_Type
ORDER BY Total_Revenue DESC;
```

RESULT

Customer Type	Total Customers	Total Orders	Total Revenue	Avg Order Value
Returning	234	648	₹273,256.40	₹421.69
New	222	641	₹265,147.95	₹413.65

KEY KPIS

- **Higher-revenue segment: Returning customers**
- **Revenue gap: ₹8,108.45**
- **Returning customers place a higher average order value**

BUSINESS INSIGHT

Of the 456 customers who have ordered at least once, 234 are returning and 222 are new — a near-even split. Returning customers still generate slightly more total revenue and a higher average order value, confirming that retained customers are marginally more valuable per order.

RECOMMENDATION

- Strengthen loyalty and repeat-purchase incentives to protect the returning-customer revenue edge.
- Personalize offers for returning customers using prior purchase history.
- Track Customer Lifetime Value (CLV) by segment to quantify the long-term payoff of retention spend versus acquisition spend.

BUSINESS VALUE

Segment-level revenue analysis helps balance acquisition and retention budgets and clarifies which segment deserves the next incremental marketing rupee.

3.4 — Which customers purchase most frequently, and who are the highest-value customers?

OBJECTIVE

Identify the customers with the highest order frequency and highest total spend to support loyalty and VIP retention programs.

SQL QUERY

```
SELECT TOP 10
    c.Customer_Name,
    c.City,
    c.Customer_Type,
    COUNT(*) AS Total_Orders,
    ROUND(SUM(o.Order_Value), 2) AS Total_Revenue,
    ROUND(AVG(o.Order_Value), 2) AS Avg_Order_Value
FROM Customers AS c
INNER JOIN Orders AS o
    ON c.Customer_ID = o.Customer_ID
GROUP BY c.Customer_Name, c.City, c.Customer_Type
ORDER BY Total_Orders DESC;
```

RESULT

Customer	City	Type	Total Orders	Total Revenue	Avg Order Value
Customer_311	Ghaziabad	New	9	₹3,744.50	₹416.06
Customer_109	Chandigarh	Returning	8	₹2,704.50	₹338.06
Customer_488	Gurgaon	New	8	₹3,321.20	₹415.15
Customer_166	Ghaziabad	New	7	₹3,001.95	₹428.85
Customer_187	Chandigarh	New	7	₹2,310.35	₹330.05
Customer_210	Faridabad	Returning	7	₹2,679.10	₹382.73
Customer_236	Faridabad	New	7	₹2,925.25	₹417.89
Customer_293	Jaipur	New	7	₹2,888.70	₹412.67
Customer_371	Bengaluru	New	7	₹2,380.95	₹340.14
Customer_56	Delhi	New	7	₹2,190.85	₹312.98

KEY KPIS

- **Most frequent buyer: Customer_311 (9 orders)**
- **Highest single-customer revenue: Customer_311 (₹3,744.50)**
- **8 of the top 10 by frequency are New customers**

BUSINESS INSIGHT

Customer_311 leads on both frequency and total revenue, making them the single highest-value account in the dataset. Notably, New customers dominate the top-10 frequency list (8 of 10) — the opposite of what the segment-level comparison above might suggest, since a handful of highly active new customers are pulling this ranking rather than the New segment as a whole.

RECOMMENDATION

- Introduce VIP or loyalty tiers for the top decile of customers by order frequency, regardless of new/returning status.
- Send personalized reorder reminders timed to each customer's historical order cadence.
- Track whether these high-frequency new customers convert into long-term returning customers, or churn after an initial burst of activity.

BUSINESS VALUE

Identifying the platform's most active individual customers — not just segment averages — lets the business protect its highest-value relationships directly rather than only through broad segment strategy.

4. Product & Category Performance

4.1 — Which product categories have the largest product catalog?

OBJECTIVE

Analyze the distribution of products across categories to understand inventory allocation and merchandising strategy.

SQL QUERY

```
SELECT Category,
       COUNT(*) AS Total_Products
FROM Products
GROUP BY Category
ORDER BY Total_Products DESC;
```

RESULT

Category	Total Products
Vegetables	12
Snacks	12
Personal Care	12
Household	12
Fruits	12
Dairy	12
Beverages	12
Bakery	12

KEY KPIS

- Every category holds exactly 12 SKUs
- 96 products total across 8 categories
- No catalog-size skew between categories

BUSINESS INSIGHT

The catalog is deliberately balanced — every category holds exactly 12 products, so no category is under- or over-represented by SKU count alone. Any performance differences seen later in this analysis (revenue, conversion) are therefore driven by demand, not catalog imbalance.

RECOMMENDATION

- Do not use catalog size as a basis for merchandising decisions — combine with sales volume, revenue and conversion instead.
- Use this even baseline as a control when comparing category performance in later sections.

BUSINESS VALUE

Confirming catalog parity up front prevents a false conclusion later — any category-level revenue or conversion gap in this report reflects genuine demand differences, not an artifact of assortment size.

4.2 — Which products generate the highest revenue?

OBJECTIVE

Identify the highest revenue-generating products to support inventory planning and promotional strategy.

SQL QUERY

```
SELECT TOP 10
    p.Product_Name,
    p.Category,
    COUNT(*) AS Total_Orders,
    ROUND(SUM(o.Order_Value), 2) AS Total_Revenue,
    ROUND(AVG(o.Order_Value), 2) AS Avg_Order_Value
FROM Products AS p
INNER JOIN Orders AS o
    ON p.Product_ID = o.Product_ID
GROUP BY p.Product_Name, p.Category
ORDER BY Total_Revenue DESC;
```

RESULT

Product	Category	Total Orders	Total Revenue	Avg Order Value
Cleaner 2	Household	21	₹16,516.50	₹786.50
Chips 2	Snacks	19	₹14,942.40	₹786.44
Soap 2	Personal Care	19	₹14,752.50	₹776.45
Juice 2	Beverages	16	₹12,774.90	₹798.43
Orange 1	Fruits	18	₹12,374.70	₹687.48
Bread 1	Bakery	13	₹11,269.90	₹866.92
Tea 3	Beverages	15	₹11,132.80	₹742.19
Cookies 1	Bakery	14	₹10,956.00	₹782.57
Cake 2	Bakery	19	₹10,847.40	₹570.92
Tea 2	Beverages	18	₹10,626.70	₹590.37

KEY KPIS

- **Top product: Cleaner 2 (₹16,516.50)**
- **Bakery and Beverages each place 3 products in the Top 10**
- **Cleaner 2 is also the #1 most-saved product (see Theme C)**

BUSINESS INSIGHT

Cleaner 2 leads on revenue and was independently identified as the most-saved product — strong evidence that its Save for Later demand is genuinely converting rather than sitting idle. Bakery and Beverages appear repeatedly in the Top 10, reinforcing Beverages' position as the top revenue category overall.

RECOMMENDATION

- Maintain high inventory availability for Cleaner 2, Chips 2 and Soap 2 to avoid stock-outs on proven demand.

- Bundle top-10 products with complementary lower-velocity items to lift average order value.
- Feature these products in homepage and recommendation placements ahead of lower-performing SKUs.

BUSINESS VALUE

Product-level revenue analysis tells inventory and merchandising exactly where to protect stock and where to invest promotional space for the best return.

4.3 — Which product categories generate the highest revenue?

OBJECTIVE

Identify the highest revenue-generating categories to support inventory planning and supplier negotiation.

SQL QUERY

```
SELECT p.Category,
       COUNT(*) AS Total_Orders,
       ROUND(SUM(o.Order_Value), 2) AS Total_Revenue,
       ROUND(AVG(o.Order_Value), 2) AS Avg_Order_Value
FROM Products AS p
INNER JOIN Orders AS o
      ON p.Product_ID = o.Product_ID
GROUP BY p.Category
ORDER BY Total_Revenue DESC;
```

RESULT

Category	Total Orders	Total Revenue	Avg Order Value
Beverages	172	₹86,271.95	₹501.58
Household	180	₹78,661.30	₹437.01
Bakery	169	₹74,943.65	₹443.45
Personal Care	163	₹72,927.15	₹447.41
Snacks	173	₹61,790.25	₹357.17
Fruits	146	₹57,333.90	₹392.70
Dairy	143	₹54,323.05	₹379.88
Vegetables	143	₹52,153.10	₹364.71

KEY KPIS

- Highest revenue category: Beverages (₹86,271.95)
- Highest average order value: Beverages (₹501.58)
- Lowest revenue category: Vegetables (₹52,153.10)

BUSINESS INSIGHT

Beverages leads on total revenue and average order value despite Household recording the highest Save-to-Purchase conversion rate elsewhere in this report (Theme C). This confirms that category revenue depends on order volume and basket value together, not conversion rate alone.

RECOMMENDATION

- Protect inventory depth for Beverages given its outsized revenue and order-value contribution.
- Expand the Beverages assortment to capture more of this demonstrated demand.
- Investigate why Vegetables underperforms on both revenue and average order value despite an identical 12-SKU catalog size to every other category.

BUSINESS VALUE

Category-level revenue analysis directs inventory investment and supplier negotiation priority toward the categories actually driving the business, rather than treating all eight categories as equal.

4.4 — Which product performs best within each category?**OBJECTIVE**

Identify the top revenue-generating product inside each category using a window function, to support per-category merchandising decisions.

SQL QUERY

```
WITH ProductRanking AS
(
    SELECT
        p.Category,
        p.Product_Name,
        SUM(o.Order_Value) AS Total_Revenue,
        DENSE_RANK() OVER (
            PARTITION BY p.Category
            ORDER BY SUM(o.Order_Value) DESC
        ) AS Product_Rank
    FROM Products AS p
    INNER JOIN Orders AS o
        ON p.Product_ID = o.Product_ID
    GROUP BY p.Category, p.Product_Name
)
SELECT Category, Product_Name, ROUND(Total_Revenue, 2) AS Total_Revenue
FROM ProductRanking
WHERE Product_Rank = 1
ORDER BY Category;
```

RESULT

Category	Top Product	Revenue
Bakery	Bread 1	₹11,269.90
Beverages	Juice 2	₹12,774.90
Dairy	Butter 1	₹9,102.00
Fruits	Orange 1	₹12,374.70
Household	Cleaner 2	₹16,516.50
Personal Care	Soap 2	₹14,752.50

Category	Top Product	Revenue
Snacks	Chips 2	₹14,942.40
Vegetables	Potato 2	₹9,979.20

KEY KPIS

- **Highest single category-leader: Cleaner 2 (Household, ₹16,516.50)**
- **Lowest category-leader: Butter 1 (Dairy, ₹9,102.00)**
- **DENSE_RANK() used to rank products within each category partition**

BUSINESS INSIGHT

Every category has one clear leading product. Cleaner 2, Soap 2 and Chips 2 — already identified as three of the platform's top-10 revenue products overall — also happen to be their category's individual leader, which reinforces confidence in their standing rather than being a coincidence of the ranking method.

RECOMMENDATION

- Guarantee stock availability for each category leader before any other SKU in that category.
- Use these eight products as anchor items in category landing pages.
- Study what makes each leader outperform its 11 category peers (price point, brand trust, pack size) and test replicating that pattern on underperformers.

BUSINESS VALUE

Per-category ranking — rather than one platform-wide Top 10 — ensures every category gets a clear merchandising priority instead of smaller categories being invisible next to platform-wide bestsellers.

5. Save for Later Engagement & Conversion

5.1 — Which products are saved by customers the most?

OBJECTIVE

Identify products with the highest customer interest to support promotional campaigns and improve purchase conversion.

SQL QUERY

```
SELECT TOP 10
    p.Product_Name,
    p.Category,
    COUNT(*) AS Total_Saves
FROM Products AS p
INNER JOIN Save_For_Later AS s
    ON p.Product_ID = s.Product_ID
GROUP BY p.Product_Name, p.Category
ORDER BY Total_Saves DESC;
```

RESULT

Product	Category	Total Saves
Cake 3	Bakery	46
Cleaner 2	Household	44
Cheese 1	Dairy	43
Cookies 3	Bakery	43
Biscuits 1	Snacks	42
Soap 2	Personal Care	42
Coffee 1	Beverages	41
Apple 3	Fruits	39
Mop 2	Household	39
Cake 2	Bakery	38

KEY KPIS

- **Most-saved product: Cake 3 (46 saves)**
- **Bakery places 3 products in the Top 10 most-saved**
- **Cleaner 2 is both a top-saved and top-revenue product**

BUSINESS INSIGHT

Cake 3 attracts the most save actions, followed closely by Cleaner 2 and Cheese 1. High save counts signal strong browsing interest, but as Theme C's conversion analysis below shows, interest alone does not guarantee a completed purchase.

RECOMMENDATION

- Layer promotions — reminders, bundles, limited-time pricing — onto the top-saved list to help intent convert into revenue.
- Compare each product's save rank against its conversion rank (see next two questions) before assuming high saves mean high value.

BUSINESS VALUE

Understanding what customers save most tells marketing where purchase intent already exists, making it a cheaper conversion target than trying to generate new demand from scratch.

5.2 — What is the overall Save-to-Purchase Conversion Rate?

OBJECTIVE

Measure the percentage of saved products that are eventually purchased, to evaluate the effectiveness of the Save for Later feature.

SQL QUERY

```
SELECT
  COUNT(*) AS Total_Saves,
  SUM(CASE WHEN Purchase_Status = 'Yes' THEN 1 ELSE 0 END) AS Total_Purchased,
  ROUND(
    SUM(CASE WHEN Purchase_Status = 'Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*),
    2) AS Conversion_Rate
FROM Save_For_Later;
```

RESULT

Total Saves	Total Purchased	Conversion Rate
3,000	1,289	42.97%

KEY KPIS

- **Total Saved Products: 3,000**
- **Purchased Products: 1,289**
- **Save-to-Purchase Conversion Rate: 42.97%**

BUSINESS INSIGHT

Just under 43% of everything saved to the list is eventually purchased, meaning the majority — 57% — is saved and never bought. That is not a failure of the feature (customers are demonstrating real purchase intent by saving at all) but it is the single largest conversion gap in this entire analysis.

RECOMMENDATION

- Send reminder notifications for items sitting in the saved list beyond a set number of days.
- Notify customers of price drops on saved products.
- Offer limited-time discounts on frequently-saved, low-converting items (see the dedicated question below).
- Treat this 42.97% figure as the platform's primary Save for Later health metric and track it over time.

BUSINESS VALUE

This is the single most important KPI for evaluating whether Save for Later is actually driving incremental revenue, and it should anchor every future iteration of the feature.

5.3 — Which product categories have the highest Save-to-Purchase Conversion Rate?

OBJECTIVE

Evaluate how effectively customer interest converts into purchases across categories.

SQL QUERY

```
SELECT p.Category,
       COUNT(*) AS Total_Saves,
       SUM(CASE WHEN s.Purchase_Status = 'Yes' THEN 1 ELSE 0 END) AS Total_Purchased,
       ROUND(
         SUM(CASE WHEN s.Purchase_Status = 'Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*),
         2) AS Conversion_Rate
FROM Products AS p
INNER JOIN Save_For_Later AS s
  ON p.Product_ID = s.Product_ID
GROUP BY p.Category
ORDER BY Conversion_Rate DESC;
```

RESULT

Category	Total Saves	Total Purchased	Conversion Rate
Household	385	180	46.75%
Snacks	387	173	44.70%
Personal Care	377	163	43.24%
Beverages	408	172	42.16%
Vegetables	340	143	42.06%
Fruits	350	146	41.71%
Dairy	343	143	41.69%
Bakery	410	169	41.22%

KEY KPIS

- **Highest conversion: Household (46.75%)**
- **Lowest conversion: Bakery (41.22%)**
- **Spread between highest and lowest: 5.53 points**

BUSINESS INSIGHT

Bakery has the most total saves of any category (410) but the lowest conversion rate — customers show the most interest here yet convert it the least. Household shows the opposite pattern: solid save volume with the platform's best conversion, suggesting essential, planned-purchase items convert more reliably than discretionary ones.

RECOMMENDATION

- Run targeted A/B tests on Bakery promotions specifically, since it has the largest untapped save volume of any category.

- Surface Bakery items during cart review and checkout as a conversion nudge.
- Investigate freshness, delivery-window or pricing concerns that might be specific to Bakery purchase hesitation.
- Study what Household is doing right and test similar messaging on Snacks and Personal Care, its closest performers.

BUSINESS VALUE

Category-level conversion analysis lets marketing prioritize the categories with the largest gap between interest and purchase, rather than spreading promotional budget evenly across all eight.

5.4 — Which products have high customer interest but low purchase conversion?

OBJECTIVE

Identify high-save, low-conversion products to uncover specific revenue optimization opportunities.

SQL QUERY

```
SELECT TOP 10
    p.Product_Name,
    p.Category,
    COUNT(*) AS Total_Saves,
    SUM(CASE WHEN s.Purchase_Status = 'Yes' THEN 1 ELSE 0 END) AS Total_Purchased,
    ROUND(
        SUM(CASE WHEN s.Purchase_Status = 'Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*),
        2) AS Conversion_Rate
FROM Products AS p
INNER JOIN Save_For_Later AS s
    ON p.Product_ID = s.Product_ID
GROUP BY p.Product_Name, p.Category
HAVING COUNT(*) >= 20
ORDER BY Conversion_Rate ASC;
```

RESULT

Product	Category	Total Saves	Total Purchased	Conversion Rate
Tea 1	Beverages	31	5	16.13%
Onion 3	Vegetables	21	4	19.05%
Milk 3	Dairy	23	5	21.74%
Yogurt 2	Dairy	31	7	22.58%
Namkeen 2	Snacks	26	6	23.08%
Bun 2	Bakery	32	8	25.00%
Shampoo 1	Personal Care	26	7	26.92%
Orange 2	Fruits	23	7	30.43%
Bread 2	Bakery	32	10	31.25%
Popcorn 3	Snacks	31	10	32.26%

KEY KPIS

- **Lowest converting product: Tea 1 (16.13%)**
- **Tea 1 has 31 saves but only 5 purchases**
- **HAVING COUNT(*) >= 20 filter excludes low-sample noise**

BUSINESS INSIGHT

Tea 1 converts at just 16.13% — the platform's weakest performer despite meaningful save volume, well below both the platform average (42.97%) and its own category average for Beverages (42.16%). A HAVING filter of at least 20 saves was applied so this ranking reflects a real pattern rather than a product with only 2-3 saves.

RECOMMENDATION

- Send targeted price-drop or limited-time-offer notifications specifically for Tea 1 and the other products on this list.
- Feature these products in personalized recommendation slots to re-surface the interest that already exists.
- Review pricing and competitor availability for Tea 1, since a 16% conversion rate against a 42% platform average points to a specific barrier, not general disengagement.

BUSINESS VALUE

This is the most actionable list in the entire report — every product on it has already proven demand through the save action, so improving its conversion requires no new customer acquisition, only removing a purchase barrier.

5.5 — Which customer segment has the highest Save-to-Purchase Conversion Rate?**OBJECTIVE**

Compare conversion rates between New and Returning customers to evaluate whether loyalty affects Save for Later effectiveness.

SQL QUERY

```
SELECT c.Customer_Type,
       COUNT(DISTINCT c.Customer_ID) AS Total_Customers,
       COUNT(*) AS Total_Saves,
       SUM(CASE WHEN s.Purchase_Status = 'Yes' THEN 1 ELSE 0 END) AS Total_Purchased,
       ROUND(
         SUM(CASE WHEN s.Purchase_Status = 'Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*),
         2) AS Conversion_Rate
FROM Customers AS c
INNER JOIN Save_For_Later AS s
  ON c.Customer_ID = s.Customer_ID
GROUP BY c.Customer_Type
ORDER BY Conversion_Rate DESC;
```

RESULT

Customer Type	Total Customers	Total Saves	Total Purchased	Conversion Rate
Returning	252	1,473	648	43.99%
New	247	1,527	641	41.98%

KEY KPIS

- **Highest conversion: Returning customers (43.99%)**
- **New customers save more in total (1,527 vs 1,473)**
- **Conversion gap: 2.01 points**

BUSINESS INSIGHT

Returning customers convert saved items slightly more often than new customers, consistent with the revenue pattern in Theme A. New customers save more in absolute terms, so even a small conversion gap represents a meaningful amount of unconverted intent.

RECOMMENDATION

- Extend the loyalty-program strategy from Theme A to specifically reinforce Save for Later engagement for returning customers.
- Test first-purchase incentives aimed at converting new customers' saved items faster, since they save more but convert less.
- Segment reminder-notification campaigns by customer type rather than sending one generic message to everyone.

BUSINESS VALUE

Segmenting conversion by customer type shows the Save for Later feature is not equally effective for every customer group, giving marketing a specific lever — new-customer conversion — to pull without touching the returning-customer experience.

6. Revenue & Payment Behavior

6.1 — Which payment methods generate the highest revenue and average order value?

OBJECTIVE

Analyze payment method performance to understand customer preferences and inform payment strategy.

SQL QUERY

```
SELECT Payment_Method,  
       COUNT(*) AS Total_Orders,  
       ROUND(SUM(Order_Value), 2) AS Total_Revenue,  
       ROUND(AVG(Order_Value), 2) AS Avg_Order_Value  
FROM Orders  
GROUP BY Payment_Method  
ORDER BY Total_Revenue DESC;
```

RESULT

Payment Method	Total Orders	Total Revenue	Avg Order Value
UPI	348	₹146,005.95	₹419.56
Debit Card	327	₹135,499.95	₹414.37
COD	312	₹132,273.40	₹423.95
Credit Card	302	₹124,625.05	₹412.67

KEY KPIS

- Highest revenue method: UPI (₹146,005.95)
- Highest average order value: COD (₹423.95)
- Most-used method by order count: UPI (348 orders)

BUSINESS INSIGHT

UPI leads on total revenue purely through order volume — it's the most-used payment method — while COD, despite fewer orders, edges out every digital method on average order value. All four methods are within a narrow ₹21,000 revenue band, so no single method dominates the business.

RECOMMENDATION

- Continue promoting UPI through cashback and digital-incentive campaigns, since it's already the customer favorite.
- Investigate why COD baskets run larger — it may indicate lower price sensitivity or different product mix — and test whether that behavior can be encouraged on digital payment methods.
- Monitor transaction costs across all four methods before shifting promotional weight toward any one of them.

BUSINESS VALUE

Payment method analysis supports checkout experience decisions and payment-partner negotiations by showing which methods actually drive revenue versus just order count.

7. Customer Segmentation

7.1 — How can customers be segmented based on revenue contribution?

OBJECTIVE

Classify customers into High, Medium and Low value tiers to support CRM strategy and personalized marketing.

SQL QUERY

```
WITH CustomerSegment AS
(
    SELECT
        c.Customer_Name,
        SUM(o.Order_Value) AS Total_Revenue,
        CASE
            WHEN SUM(o.Order_Value) >= 3000 THEN 'High'
            WHEN SUM(o.Order_Value) >= 1500 THEN 'Medium'
            ELSE 'Low'
        END AS Customer_Segment
    FROM Customers AS c
    INNER JOIN Orders AS o
        ON c.Customer_ID = o.Customer_ID
    GROUP BY c.Customer_Name
)
SELECT
    Customer_Segment,
    COUNT(*) AS Total_Customers,
    ROUND(SUM(Total_Revenue), 2) AS Total_Revenue,
    ROUND(AVG(Total_Revenue), 2) AS Avg_Revenue
FROM CustomerSegment
GROUP BY Customer_Segment
ORDER BY Avg_Revenue DESC;
```

RESULT

Segment	Total Customers	Total Revenue	Avg Revenue / Customer
High	10	₹32,372.65	₹3,237.27
Medium	123	₹251,102.70	₹2,041.49
Low	323	₹254,929.00	₹789.25

KEY KPIS

- **High-Value customers: 10 (2.2% of ordering customers)**
- **Medium-Value customers: 123 (27.0%)**
- **Low-Value customers: 323 (70.8%), yet contribute the most total revenue**

BUSINESS INSIGHT

The customer base is heavily weighted toward Low Value — over 70% of ordering customers fall into this tier, and their sheer number means they collectively out-earn the Medium tier in total revenue despite spending far less

individually. The 123 Medium-Value customers represent the clearest growth lever: fewer than half as numerous as the Low tier, but nearly triple the per-customer revenue.

RECOMMENDATION

- Protect the 10 High-Value customers with VIP treatment, premium support and exclusive early access.
- Prioritize Medium-Value customers for upsell and bundling campaigns — this tier has the best ratio of headcount to growth potential.
- Use low-cost, high-reach tactics (coupons, re-engagement emails) for the Low tier rather than high-touch campaigns, given its size and lower per-customer return.

BUSINESS VALUE

Segmenting by revenue contribution — rather than treating all 456 ordering customers identically — lets the business match campaign cost to expected return per tier.

8. Time-Based & Advanced Analysis (New in v2.0)

8.1 — How has revenue trended month over month across the analysis period?

OBJECTIVE

Track monthly order volume and revenue to identify growth, seasonality, or decline patterns the point-in-time metrics above cannot show.

SQL QUERY

```
SELECT
    FORMAT(Order_Date, 'yyyy-MM') AS Order_Month,
    COUNT(*) AS Total_Orders,
    ROUND(SUM(Order_Value), 2) AS Total_Revenue
FROM Orders
GROUP BY FORMAT(Order_Date, 'yyyy-MM')
ORDER BY Order_Month;
```

RESULT

Month	Total Orders	Total Revenue
2026-06	118	₹53,579.50
2026-07	247	₹102,286.40
2026-08	202	₹77,600.90
2026-09	197	₹86,172.65
2026-10	220	₹91,414.10
2026-11	228	₹94,456.25
2026-12	77	₹32,894.55

KEY KPIS

- **Peak month: July 2026 (₹102,286.40)**
- **Lowest month: December 2026 (₹32,894.55, partial month in the dataset)**
- **5 of 7 months show revenue above ₹75,000**

BUSINESS INSIGHT

Revenue peaks sharply in July before settling into a steadier ₹77K-₹94K band from August through November. December's figure is the lowest in the dataset, but the order dates only run through mid-December, so this is a partial month rather than a genuine decline — a caveat worth stating explicitly rather than reading as a drop-off.

RECOMMENDATION

- Investigate what drove the July spike (promotion, seasonal demand, or a data generation artifact) before using it as a planning baseline.
- Treat December as an incomplete month in any month-over-month comparison to avoid an inaccurate 'December decline' conclusion.
- Extend the observation window in a future data refresh to properly evaluate seasonality across a full 12-month cycle.

BUSINESS VALUE

Monthly trending turns a set of static totals into a timeline, which is what leadership actually needs to plan inventory, staffing and campaign timing — a single 'Total Revenue' figure cannot support that decision on its own.

8.2 — Which day of the week drives the most orders and revenue?

OBJECTIVE

Identify weekday-level demand patterns to support staffing, delivery capacity, and promotional timing decisions.

SQL QUERY

```
SELECT
    DATENAME(WEEKDAY, Order_Date) AS Day_Of_Week,
    COUNT(*) AS Total_Orders,
    ROUND(SUM(Order_Value), 2) AS Total_Revenue,
    ROUND(AVG(Order_Value), 2) AS Avg_Order_Value
FROM Orders
GROUP BY DATENAME(WEEKDAY, Order_Date), DATEPART(WEEKDAY, Order_Date)
ORDER BY Total_Revenue DESC;
```

RESULT

Day	Total Orders	Total Revenue	Avg Order Value
Monday	198	₹87,770.70	₹443.29
Thursday	188	₹78,353.70	₹416.77
Friday	190	₹78,185.75	₹411.50
Tuesday	181	₹77,510.90	₹428.24
Saturday	195	₹76,470.15	₹392.15
Wednesday	160	₹72,066.60	₹450.42
Sunday	177	₹68,046.55	₹384.44

KEY KPIS

- **Highest-revenue day: Monday (₹87,770.70)**
- **Highest average order value: Wednesday (₹450.42)**
- **Lowest-revenue day: Sunday (₹68,046.55)**

BUSINESS INSIGHT

Monday generates the most orders and the most revenue of any day, while Sunday is consistently the weakest — a pattern worth testing against, since grocery delivery demand often skews toward weekends in other markets. Wednesday has the fewest orders of any weekday but the highest average order value, suggesting a smaller number of more deliberate, larger baskets.

RECOMMENDATION

- Schedule promotional pushes and delivery capacity around the Monday peak rather than assuming a weekend-heavy pattern.

- Test a Sunday-specific incentive (free delivery, limited-time discount) to lift the platform's weakest day.
- Investigate what drives Wednesday's larger basket size and whether it can be replicated on lower-value days.

BUSINESS VALUE

Day-of-week analysis lets operations align staffing and delivery capacity with actual demand rather than assumed patterns, and gives marketing a specific day to target with a re-engagement offer.

8.3 — What is the month-over-month revenue growth rate?

OBJECTIVE

Quantify growth or decline between consecutive months using a window function, to complement the raw monthly totals above.

SQL QUERY

```
WITH Monthly AS (
  SELECT
    FORMAT(Order_Date, 'yyyy-MM') AS Order_Month,
    ROUND(SUM(Order_Value), 2) AS Total_Revenue
  FROM Orders
  GROUP BY FORMAT(Order_Date, 'yyyy-MM')
)
SELECT
  Order_Month,
  Total_Revenue,
  LAG(Total_Revenue) OVER (ORDER BY Order_Month) AS Prev_Month_Revenue,
  ROUND(
    (Total_Revenue - LAG(Total_Revenue) OVER (ORDER BY Order_Month)) * 100.0
    / LAG(Total_Revenue) OVER (ORDER BY Order_Month),
    2) AS MoM_Growth_Pct
FROM Monthly
ORDER BY Order_Month;
```

RESULT

Month	Revenue	Prev Month Revenue	MoM Growth %
2026-06	₹53,579.50	—	—
2026-07	₹102,286.40	₹53,579.50	+90.91%
2026-08	₹77,600.90	₹102,286.40	-24.13%
2026-09	₹86,172.65	₹77,600.90	+11.05%
2026-10	₹91,414.10	₹86,172.65	+6.08%
2026-11	₹94,456.25	₹91,414.10	+3.33%
2026-12	₹32,894.55	₹94,456.25	-65.17%

KEY KPIS

- **Largest month-over-month gain: July (+90.91%)**
- **Largest month-over-month drop: December (-65.17%, partial month)**
- **3 consecutive months of positive growth: September-November**

BUSINESS INSIGHT

After the June-to-July surge, the business shows a sharp August correction followed by three consecutive months of steady, moderating growth (September through November) — a healthier and more sustainable pattern than the initial spike. December's apparent -65% drop should be read alongside the previous question's caveat: the month is incomplete in this dataset, not a genuine collapse.

RECOMMENDATION

- Use September-November's steady single-digit growth, not July's spike, as the realistic baseline for forecasting.
- Exclude the partial December month from any growth-rate calculation used in planning or reporting.
- Investigate the August correction specifically — a 24% drop immediately after a 91% spike suggests the July number may include a one-time event worth understanding.

BUSINESS VALUE

A single 'Total Revenue' KPI hides exactly this kind of volatility. Month-over-month growth analysis using LAG() gives leadership the trend line, not just the total, which is what's actually needed to judge whether the business is accelerating, stabilizing, or declining.

8.4 — What does cumulative revenue look like across the analysis period?

OBJECTIVE

Build a running revenue total using a window function to show cumulative growth toward the ₹538,404.35 total reported elsewhere in this analysis.

SQL QUERY

```
WITH Monthly AS (
    SELECT
        FORMAT(Order_Date, 'yyyy-MM') AS Order_Month,
        ROUND(SUM(Order_Value), 2) AS Total_Revenue
    FROM Orders
    GROUP BY FORMAT(Order_Date, 'yyyy-MM')
)
SELECT
    Order_Month,
    Total_Revenue,
    ROUND(
        SUM(Total_Revenue) OVER (
            ORDER BY Order_Month
            ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
        ),
        2) AS Running_Total_Revenue
FROM Monthly
ORDER BY Order_Month;
```

RESULT

Month	Monthly Revenue	Running Total
2026-06	₹53,579.50	₹53,579.50
2026-07	₹102,286.40	₹155,865.90
2026-08	₹77,600.90	₹233,466.80
2026-09	₹86,172.65	₹319,639.45
2026-10	₹91,414.10	₹411,053.55
2026-11	₹94,456.25	₹505,509.80
2026-12	₹32,894.55	₹538,404.35

KEY KPIS

- Running total reconciles exactly to ₹538,404.35 — the platform's reported Total Revenue
- 50% of total revenue was earned by end of September
- First two months (Jun-Jul) contributed 29% of the full period's revenue

BUSINESS INSIGHT

The running total reconciles precisely with the Total Revenue figure reported in the Executive KPI Summary, confirming the monthly breakdown is internally consistent with every other total in this analysis. The business crossed the halfway point of its total revenue by the end of September, roughly two-thirds of the way through the observed period.

RECOMMENDATION

- Use this running-total view as the standard way to track progress toward a revenue target inside a reporting period, rather than only showing a single end-of-period total.
- Pair this cumulative view with the month-over-month growth table above when presenting to stakeholders — one shows pace, the other shows position.

BUSINESS VALUE

A running total is the standard way analysts show progress-to-goal; it lets a reader see at a glance whether the business is ahead of, on, or behind pace without doing month-by-month subtraction themselves.

8.5 — Within each city, which customers spend above their city's own average?**OBJECTIVE**

Identify above-average spenders on a per-city basis using a correlated subquery, since a single platform-wide average would unfairly flag customers in high-cost cities as low performers.

SQL QUERY

```
SELECT c.City, c.Customer_ID,
       ROUND(SUM(o.Order_Value), 2) AS Customer_Revenue
FROM Customers AS c
INNER JOIN Orders AS o
      ON c.Customer_ID = o.Customer_ID
GROUP BY c.City, c.Customer_ID
HAVING SUM(o.Order_Value) > (
```

```

SELECT AVG(city_totals.Cust_Total)
FROM (
    SELECT o2.Customer_ID, SUM(o2.Order_Value) AS Cust_Total
    FROM Customers AS c2
    INNER JOIN Orders AS o2
        ON c2.Customer_ID = o2.Customer_ID
    WHERE c2.City = c.City
    GROUP BY o2.Customer_ID
) AS city_totals
)
ORDER BY c.City, Customer_Revenue DESC;

```

RESULT

City	Ordering Customers	Above City Average	City Avg Revenue / Customer
Faridabad	51	23	₹1,165.88
Pune	51	23	₹1,084.70
Bengaluru	44	22	₹1,238.71
Delhi	45	22	₹1,154.02
Ghaziabad	49	20	₹1,346.14
Jaipur	47	19	₹1,212.11
Lucknow	43	19	₹1,078.25
Chandigarh	48	16	₹1,181.89
Gurgaon	40	15	₹1,277.91
Noida	38	15	₹1,053.90

KEY KPIS

- 217 of 456 ordering customers (47.6%) spend above their own city's average
- Ghaziabad has the highest city average (₹1,346.14) despite only 20 of 49 customers clearing it
- Result set correlates the subquery to each outer row's City, recalculating the benchmark per city

BUSINESS INSIGHT

This query recalculates the average separately for every city rather than using one platform-wide number, so a ₹1,200 spender in Noida (above its ₹1,053.90 average) and a ₹1,200 spender in Ghaziabad (below its ₹1,346.14 average) are correctly told apart. Roughly 48% of ordering customers clear their own city's bar — a more meaningful signal than comparing every customer to a single national average.

RECOMMENDATION

- Use city-relative spend, not a flat platform-wide average, when qualifying customers for city-level loyalty or upgrade offers.
- Prioritize the below-average majority in each city (52.4%) for spend-lift campaigns, since they represent the larger, more addressable group.

- Revisit this segmentation quarterly as city averages shift, since a customer's relative standing can change even if their own spend doesn't.

BUSINESS VALUE

A correlated subquery lets each city set its own bar, which is a fairer and more actionable way to identify above-average customers than a single company-wide threshold that structurally favors customers in already-high-spending cities.

9. Executive KPI Summary

9.1 — What are the company's key business KPIs from an executive perspective?

OBJECTIVE

Consolidate the platform's headline metrics into a single executive-level view of customer activity, revenue, and Save for Later performance.

SQL QUERY

```
SELECT
  (SELECT COUNT(*) FROM Customers) AS Total_Customers,
  (SELECT COUNT(*) FROM Orders) AS Total_Orders,
  (SELECT ROUND(SUM(Order_Value), 2) FROM Orders) AS Total_Revenue,
  (SELECT ROUND(AVG(Order_Value), 2) FROM Orders) AS Avg_Order_Value,
  (SELECT ROUND(SUM(Order_Value) * 1.0 / (SELECT COUNT(*) FROM Customers), 2)
   FROM Orders) AS Revenue_Per_Customer,
  (SELECT COUNT(*) FROM Save_For_Later) AS Total_Saves,
  (SELECT ROUND(
    SUM(CASE WHEN Purchase_Status = 'Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*),
    2) FROM Save_For_Later) AS Save_To_Purchase_Conversion;
```

RESULT

Metric	Value
Total Customers	500
Total Orders	1,289
Total Revenue	₹538,404.35
Average Order Value	₹417.69
Revenue Per Customer	₹1,076.81
Total Saved Products	3,000
Save-to-Purchase Conversion Rate	42.97%

KEY KPIS

- **Total Revenue: ₹538,404.35**
- **Revenue Per Customer: ₹1,076.81**
- **Save-to-Purchase Conversion: 42.97%**

BUSINESS INSIGHT

The platform shows healthy overall performance: consistent average order values, an engaged customer base, and a Save for Later feature that converts nearly 43% of saved intent into completed purchases. Revenue Per Customer here is calculated against the full 500-customer base (registered, not just ordering customers) to align exactly with the Power BI Executive Dashboard's definition of the metric — see the Data Notes section for why that denominator was chosen.

RECOMMENDATION

- Track these seven metrics as the platform's standing executive dashboard, refreshed on a consistent cadence.

- Prioritize Save-to-Purchase Conversion improvement, since it is the metric with the most direct, immediately actionable growth path shown throughout this report.
- Report Revenue Per Customer using the same registered-customer denominator across every artifact (dashboard, this document, any future report) to avoid the inconsistency this report's v1.0 originally contained.

BUSINESS VALUE

A single consolidated KPI view gives leadership a one-glance read on business health without requiring them to review every underlying question in this document.

10. Strategic Synthesis

Based on the complete analysis, what is the biggest business opportunity for the company?

ANALYSIS SUMMARY

- Save-to-Purchase Conversion Rate stands at 42.97% — 57% of saved products are never purchased.
- Returning customers generate more revenue and convert saved items slightly better than new customers.
- Several individual products (Tea 1, Onion 3, Milk 3) show strong save volume but conversion well below the platform average.
- Beverages leads on revenue while Household leads on conversion — category performance depends on both volume and effectiveness.
- Medium-Value customers (123 of 456 ordering customers) represent the largest realistic upgrade path to High-Value status.
- Revenue is not flat across the year — July's spike and the Aug-Nov steady growth band suggest real seasonality worth planning around.

BUSINESS INSIGHT

The single largest opportunity is closing the Save-to-Purchase conversion gap. Customers are already demonstrating purchase intent by saving 3,000 products — the business does not need to acquire new demand, it needs to convert demand that already exists. This is consistently cheaper than new customer acquisition and is supported by evidence at the platform, category and individual-product level throughout this report.

RECOMMENDATION

- Implement price-drop notifications for saved products.
- Notify customers when a saved, previously out-of-stock product returns to availability.
- Send personalized reminders for saved items approaching 14+ days without purchase.
- Recommend complementary products and bundles at checkout for saved items.
- Launch limited-time offers targeted specifically at the lowest-converting high-save products identified in this report (Tea 1, Onion 3, Milk 3, Yogurt 2).
- Focus retention spend on Medium-Value customers to grow their progression into the High-Value tier.

EXPECTED BUSINESS IMPACT

- Higher Save-to-Purchase Conversion Rate
- Increased Average Order Value through bundling
- Improved customer retention and repeat purchase rate
- Higher overall platform revenue without a proportional increase in acquisition cost

BUSINESS VALUE

By focusing on customers who have already shown purchase intent through the Save for Later feature, the business can pursue sustainable revenue growth at a lower cost than acquisition-led strategies — while directly improving the metric (Save-to-Purchase Conversion) that this entire analysis identifies as the platform's biggest single lever.

Analysis performed in Microsoft SQL Server (T-SQL). All results in this document were generated by executing the queries above directly against the project's Customers, Orders, Products and Save_For_Later tables — figures are not manually estimated.